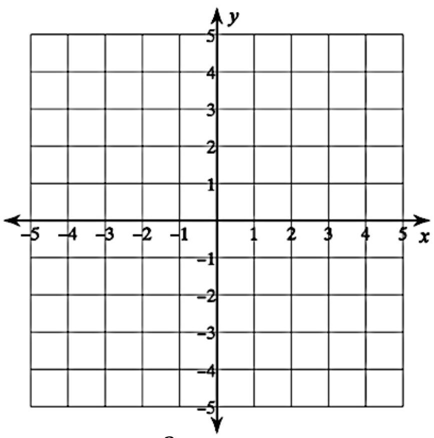
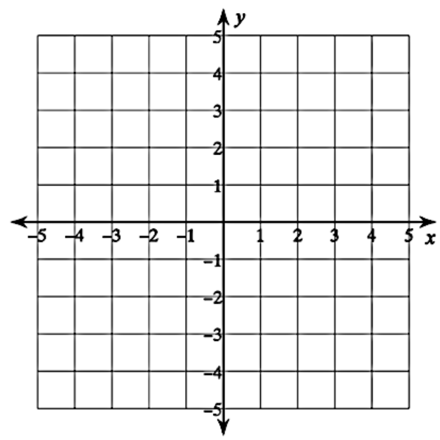
U2-D3 Homework – Absolute Value Graphs **Name:**

Graph each function. **State the vertex, slope *and D/R*.** Graphs must show vertex and at least one point on each side of the V (but more points are nicer if they fit).

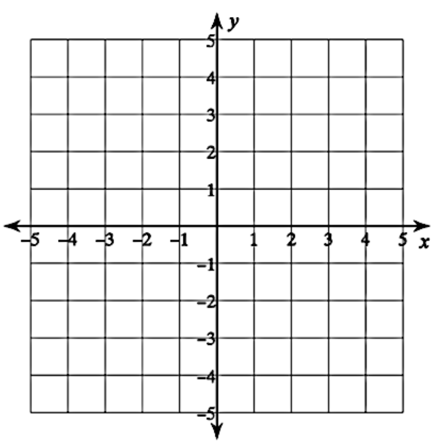
1. $f(x) = |x-2| + 1$



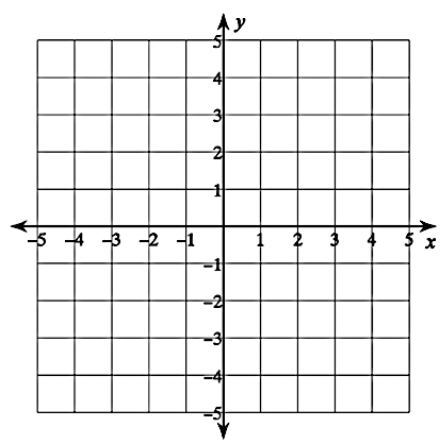
2. $y = -3|x| + 4$



3. $y = \frac{3}{2}|x+1| - 3$



4. $h(x) = 2|x+3|$



5. Solve the absolute value equation.

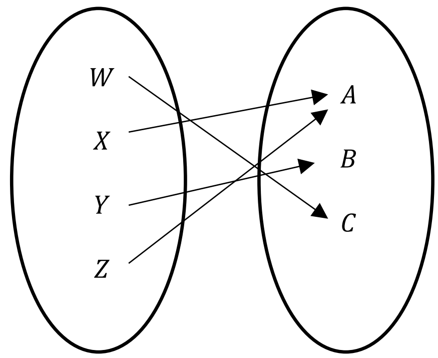
$$-2|x-1| + 23 = -7$$

Determine if each relation represents a function and state its domain and range.

6.

x	y
1	5
2	0
1	5
2	0
1	-5
2	0

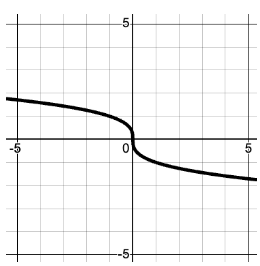
7.



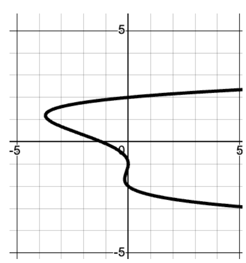
8. Express the relation from #6 as a set of ordered pairs.

Determine if each graph represents a function and state its domain and range. *If it is a function,* **state end behavior** using words written as an ordered pair.

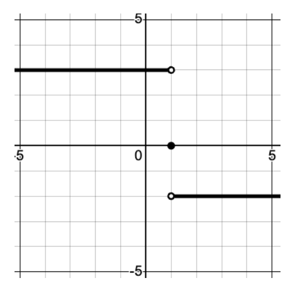
9.



10.



11.



12. In #11, what is $f(1)$? Explain.

13. Solve by **substitution**, then confirm the result using a graphing calculator.

$$2x + 3y = 3$$
$$x + 8y = -5$$